

Inhibitory effects of cranberry juice on attachment of oral streptococci and biofilm formation.



Cranberry juice is known to inhibit bacterial adhesion. We examined the inhibitory effect of cranberry juice on the adhesion of oral streptococci strains labeled with [3H]-thymidine to saliva-coated hydroxyapatite beads (s-HA). When the bacterial cells were momentarily exposed to cranberry juice, their adherence to s-HA decreased significantly compared with the control ($P < 0.01$). Their hydrophobicity also decreased dependently with the concentration of cranberry juice. We also evaluated the inhibitory effect of cranberry juice on biofilm formation. By using a microplate system, we found that the high molecular mass constituents of cranberry juice inhibited the biofilm formation of the tested streptococci. The inhibitory activity was related to the reduction of the hydrophobicity. The present findings suggest that cranberry juice component(s) can inhibit colonization by oral streptococci to the tooth surface and can thus slow development of dental plaque. Copyright Blackwell Munksgaard, 2004.